

THE (ALMOST) IMPOSSIBLE PUZZLE

ABSTRACT. Tom, Jerry and Spike play the following game. Tom generates a random binary code of length n , which he sends to Spike along with a *secret number* k between 0 and $n - 1$. Spike flips exactly one bit in the code and then sends only the modified binary code to Jerry. Is there a pre-established strategy for Spike and Jerry such that Jerry can determine the *secret number* k solely by looking at the code sent by Spike?

1. DEFINITIONS

Let $\text{Fin } n := \{0, \dots, n - 1\}$, $n \in \mathbb{N}$ ($\text{Fin } 0 := \emptyset$). $\mathbb{Z}_2 := \{0, 1\}$ is the commutative ring of integers modulo 2.

def:Z2n **Definition 1.1.** $\mathbb{Z}_2^n := \mathbb{Z}_2 \times \dots \times \mathbb{Z}_2$. $\mathbb{Z}_2^n$ can be seen as the set of all functions from $\text{Fin } n$ to $\mathbb{Z}_2$. We consider $\mathbb{Z}_2^n$ as a module over $\mathbb{Z}_2$.

def:e **Definition 1.2.** For $i \in \text{Fin } n$, let $e_i \in \mathbb{Z}_2^n$ be the canonical vector with 1 on the i -th position and zeros elsewhere

e (e)
$$e_i := (0, \dots, 0, 1, 0, \dots, 0),$$

e_i can be also seen as the function defined on $\text{Fin } n$, supported at i , with value 1 there, and 0 elsewhere.

Note that $b + e_i$ is the code b with the i -th bit flipped, for every $b \in \mathbb{Z}_2^n$ and $i \in \text{Fin } n$.

def:SF **Definition 1.3.** We say that $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is a **strategy function**, if

SF (SF)
$$\forall b \in \mathbb{Z}_2^n, k \in \text{Fin } n, \exists i \in \text{Fin } n, f(b + e_i) = k.$$

We say that the puzzle **has a strategy** for n if

$$\exists f : \mathbb{Z}_2^n \rightarrow \text{Fin } n, \text{ (SF)}.$$

def:d **Definition 1.4.** Let $\delta : \text{Fin } n \times \text{Fin } n \rightarrow \mathbb{N}$ be given by

d (d)
$$\delta(i, j) = \text{if } i = j \text{ then } 1 \text{ else } 0.$$

See the Mathlib definition [if-then-else](#).

2. MAIN RESULT

In this section, we characterize the values of n for which the puzzle has a strategy and we present a sketch of the proof. The details of the proof are given in the next sections and are adapted for formalization in Lean.

Theorem *The puzzle has a strategy for $n \in \mathbb{N}^*$ if and only if n is a power of 2.*

Sketch of the proof.

Modus Ponens For every $b \in \mathbb{Z}_2^n$,

$$\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n \text{ is surjective,}$$

and thus also injective. Fix some $k \in \text{Fin } n$ (e.g., $k = 0 < n$). Then

$$\begin{aligned} 2^n &= \sum_{b \in \mathbb{Z}_2^n} 1 = \sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = \sum_{i \in \text{Fin } n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b + e_i), k) \\ &= \sum_{i \in \text{Fin } n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = n \cdot \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k). \end{aligned}$$

Modus Ponens Reverse Let $n = 2^m$. Let $g : \text{Fin } (2^m) \rightarrow \mathbb{Z}_2^m$ be a bijection. Define

$$f(b) := g^{-1} \left(\sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right), \quad b \in \mathbb{Z}_2^n.$$

Let $b \in \mathbb{Z}_2^n$ and $k \in \text{Fin } n$.

$$\begin{aligned} f(b + e_i) = k &\Leftrightarrow \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + \sum_{j \in \text{Fin } 2^m} e_{ij} \cdot g(j) = g(k) \\ &\Leftrightarrow \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + g(i) = g(k) \Leftrightarrow i = g^{-1} \left(g(k) - \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right). \end{aligned}$$

3. LEAN TACTICS

- `apply`
- `by_contra`
- `constructor`
- `dsimp`
- `exact`
- `have`
- `intro`
- `let`
- `obtain`
- `rw`
- `simp (only)`
- `unfold`
- `use`

4. FUNCTION SHIFT IS INJECTIVE

`fun_shift_surj`

Lemma 4.1. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF) and $b \in \mathbb{Z}_2^n$, then the function $\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n$ is *surjective*.*

Proof. $\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n$ is surjective, by definition, if

$$\forall k \in \text{Fin } n, \exists i \in \text{Fin } n, f(b + e_i) = k.$$

Taking our b in the definition of (SF), we get the desired conclusion. $\square$

Mathlib hint: `Function.Surjective`.

`fun_shift_inj`

Lemma 4.2. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF) and $b \in \mathbb{Z}_2^n$, then the function $\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n$ is *injective*.*

Proof. Since $\text{Fin } n \ni i \mapsto f(b + e_i) \in \text{Fin } n$ is a *surjective* function, by Lemma 4.1, on a *finite* set, it is also *injective*. $\square$

Mathlib hints: `Function.Injective`, `Finite.injective_iff_surjective`.

5. FUNCTION SHIFT SUM

`fun_shift_sum_one`

Lemma 5.1. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF), $k \in \text{Fin } n$, and $b \in \mathbb{Z}_2^n$, then*

$$\sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = 1.$$

Proof. In view of the definition of (SF) for f , we obtain $i_k \in \text{Fin } n$ such that $f(b + e_{i_k}) = k$.

We shall *rewrite* the sum by proving that *only the i_k term in the sum is equal to 1, and the rest are 0*.

Clearly, $\delta(f(b + e_{i_k}), k) = 1$, by the definition of (d) and the logic of *if-then-else*.

Introduce $i \in \text{Fin } n$ with $i \neq i_k$ and consider the definition of (d). *Rewriting* k as $f(b + e_{i_k})$ and *applying* the logic of *if-then-else* in (d), we have to prove that $f(b + e_i) \neq f(b + e_{i_k})$. In order to get a *contradiction*, assume that $f(b + e_i) = f(b + e_{i_k})$. Then, by Lemma 4.2, we *have* $i = i_k$, which is *false* in view of the assumption $i \neq i_k$. $\square$

Mathlib hints: `rw [Fintype.sum_eq_single i_k]`, `if_pos`, `if_neg`.

`fun_shift_sum_pow2`

Lemma 5.2. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF) and $k \in \text{Fin } n$, then*

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = 2^n.$$

Proof. By Lemma 5.1, we *have*:

$$\forall b \in \mathbb{Z}_2^n, \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = 1.$$

With *only* this relation, we can *simplify* $\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k)$ to $\sum_{b \in \mathbb{Z}_2^n} 1$,

which *simplifies* further to 2^n . $\square$

fun_shift_sum_rev

Lemma 5.3. *If $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ and $k, i \in \text{Fin } n$, then*

$$\sum_{b \in \mathbb{Z}_2^n} \delta(f(b + e_i), k) = \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k).$$

Proof. Let $g : \mathbb{Z}_2^n \rightarrow \mathbb{Z}_2^n, g(b) = b + e_i$. Then we have that g is bijective, because adding a constant is both injective and surjective on $\mathbb{Z}_2^n$. Applying the fact that g is bijective to our finite sum, the equality holds, because $\sum_{b \in \mathbb{Z}_2^n} \delta(f(g(b)), k)$ is simply $\sum_{b \in \mathbb{Z}_2^n} \delta(f(b + e_i), k)$. $\square$

Mathlib hints: `Function.Bijective`, `add_left_injective`, `add_right_surjective`, `Fintype.sum_bijective`.

6. MODUS PONENS

strategy_pow2

Lemma 6.1. *If $0 < n$ and $f : \mathbb{Z}_2^n \rightarrow \text{Fin } n$ is (SF), then $\exists N \in \mathbb{N}, n \cdot N = 2^n$.*

Proof. Let $k := 0$ (which is in $\text{Fin } n$ since $0 < n$). By Lemma 5.2, we have

$$\sum_{b \in \mathbb{Z}_2^n} \sum_{i \in \text{Fin } n} \delta(f(b + e_i), k) = 2^n.$$

Since the sums commute, we can rewrite the above expression as

$$\sum_{i \in \text{Fin } n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b + e_i), k) = 2^n.$$

Using Lemma 5.3, we simplify this to have

$$\sum_{i \in \text{Fin } n} \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = n \cdot \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k) = 2^n.$$

We use $N := \sum_{b \in \mathbb{Z}_2^n} \delta(f(b), k)$ to finish the proof. $\square$

Mathlib hint: `Finset.sum_comm`.

mul_pow2

Lemma 6.2. *If $\exists N \in \mathbb{N}, n \cdot N = 2^n$, then $\exists m \in \mathbb{N}, n = 2^m$.*

Proof. From the hypothesis, we obtain N such that $n \cdot N = 2^n$. So, we have n divides 2^n . Since 2 is prime, we obtain $m \in \mathbb{N}$ to be used in $n = 2^m$. $\square$

Mathlib hints: `Dvd.intro`, `Nat.prime_two`, `Nat.dvd_prime_pow`.

puzzle_mp

Theorem 6.3. *If $0 < n$ and the puzzle has a strategy for n , then $\exists m \in \mathbb{N}, n = 2^m$.*

Proof. Since the puzzle has a strategy, we obtain f with (SF). By putting together Lemma 6.1 and Lemma 6.2, $\exists m \in \mathbb{N}, n = 2^m$. $\square$

7. MODUS PONENS REVERSE

`bij_Fin`

Definition 7.1. For $m \in \mathbb{N}$, let $g : \text{Fin } (2^m) \rightarrow \mathbb{Z}_2^m$ be a bijective function. g exists, [applying](#) the fact that $\text{Fin } (2^m)$ and $\mathbb{Z}_2^m$ [simply](#) have [equal cardinals](#).

Mathlib hint: [Fintype.equivOfCardEq](#).

`sum_basis`

Lemma 7.2. If $i \in \text{Fin } n$ and $f : \text{Fin } n \rightarrow \mathbb{Z}_2^m$, then $\sum_{j \in \text{Fin } n} e_{ij} \cdot f(j) = f(i)$.

Proof. By the definition of [\(e\)](#), $e_{ij} = 1$ if $i = j$ and $e_{ij} = 0$ otherwise. We [rewrite](#) $\sum_{j \in \text{Fin } n} e_{ij} \cdot f(j)$ as $f(i)$, using the fact that the [single](#) term that contributes in the [sum](#) is the one with $j = i$. $\square$

Mathlib hint: [Fintype.sum_single_smul](#).

`puzzle_mpr`

Theorem 7.3. If $n = 2^m$ for some m , then the puzzle has a strategy for n .

Proof. First, we [rewrite](#) n as 2^m in the definition of [\(SF\)](#). Let g be given by [Definition 7.1](#) and [use](#) for f in [\(SF\)](#) the function

$$b \mapsto g^{-1} \left(\sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right).$$

[Introduce](#) $b \in \mathbb{Z}_2^n$ and $k \in \text{Fin } n$ in order to verify the conditions of [\(SF\)](#). We want to show

$$\exists i \in \text{Fin } n, f(b + e_i) = g^{-1} \left(\sum_{j \in \text{Fin } 2^m} (b_j + e_{ij}) \cdot g(j) \right) = k.$$

We [simplify](#) the above relation using the [distributivity of the scalar multiplication over addition](#) and over [sum](#) to get

$$\sum_{j \in \text{Fin } 2^m} (b_j + e_{ij}) \cdot g(j) = \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + \sum_{j \in \text{Fin } 2^m} e_{ij} \cdot g(j).$$

Let's [use](#) $i := g^{-1} \left(g(k) - \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right)$. By [Lemma 7.2](#), we can [rewrite](#) the desired relation with $\sum_{j \in \text{Fin } 2^m} e_{ij} \cdot g(j) = g(i)$. By the expression of i , we can [simplify](#) to get

$$\begin{aligned} & g^{-1} \left(\sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + g(i) \right) \\ &= g^{-1} \left(\sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) + g(k) - \sum_{j \in \text{Fin } 2^m} b_j \cdot g(j) \right) = g^{-1}(g(k)) = k. \end{aligned}$$

 $\square$

Mathlib hints: [g.symm](#), [add_smul](#), [Finset.sum_add_distrib](#).

8. FINISH

puzzle **Theorem 8.1.** *If $0 < n$, then the puzzle has a strategy for n if and only if $\exists m \in \mathbb{N}, n = 2^m$.*

Proof. The forward direction is [Theorem 6.3](#), and the reverse direction is [Theorem 7.3](#). □